

PAPER_849: Integrated 24-Paper arXiv BSM Landscape — F_quark, F_neutrino, F_ALP, F_dark UQFF Bridge

Author: Daniel T. Murphy | **Framework:** UQFF v5.57 **Session:** 197 | **Date:** June 19-20, 2025 **Shares:** UQFF_arXivChandra_20250619_0803PM, UQFF_arXivChandra2_20250619_0947PM, UQFF_arXivChandra3_20250620_0735AM, UQFF_arXivChandra4_20250620_0803AM

Abstract

Twenty-four arXiv publications from high-energy particle physics are analyzed across four batches to derive four beyond-Standard-Model (BSM) $F_{U_{Bi}i}$ coupling terms. F_{quark} dominates at $1.54e7$ N from CKM matrix element $|V_{cb}| = 39.2e-3$ (Belle II). F_{neutrino} (1.0 N), F_{ALP} ($1e4$ N), and F_{dark} (1.0 N) contribute subdominant but physically meaningful couplings connecting particle physics to UQFF's unified buoyancy framework.

1. Batch 1 — F_quark (5 papers)

$$F_{\text{quark}} = k_{\text{quark}} * |V_{cb}|^2 = 10^{10} * (39.2e-3)^2 = 1.54e7 \text{ N}$$

Primary: arXiv:2506.15256 (Belle II $B \rightarrow D^*$ exclusive $|V_{cb}|$)

Also: 2506.15347, 2506.15390, 2506.15515, 2506.15533

$|V_{cb}| = 39.2 \pm 0.7 \times 10^{-3}$ (Belle II 2025 average)

CKM unitarity: $\text{sum of } |V_{ub}|^2 + |V_{cb}|^2 + |V_{tb}|^2 = 1$

2. Batch 2 — F_neutrino (6 papers)

$$F_{\text{neutrino}} = k_{\text{neutrino}} * \alpha_{\nu} = 10^{10} * 10^{-10} = 1.0 \text{ N}$$

Primary: arXiv:2506.14881 (neutrino electromagnetic polarizability)

Also: 2506.14989, 2506.15046, 2506.15164, 2506.15245, 2506.15306

$\alpha_{\nu} = 10^{-10}$ (neutrino polarizability upper bound)

Connects neutrino mass generation to vacuum coupling.

3. Batch 3 — F_ALP (6 papers)

$$F_{\text{ALP}} = k_{\text{ALP}} * g_{\text{aqq}} = 10^{10} * 10^{-6} = 10^4 \text{ N}$$

Primary: arXiv:2506.15637 (ALP-hadron covariant derivative coupling)

Also: 2506.15428, 2506.15445, 2412.04357, 2412.10141, 2503.05679

$g_{\text{aqq}} = 10^{-6}$ (ALP-quark coupling constant)

ALPs as dark matter candidates with vacuum energy coupling.

MicroLHC beam-dump sensitivity: $g_{\text{aqq}} < 10^{-5}$ projected.

4. Batch 4 — F_dark (7 papers)

$$F_{\text{dark}} = k_{\text{dark}} * \epsilon^2 = 10^{10} * (10^{-5})^2 = 1.0 \text{ N}$$

Primary: arXiv:2402.00249 (FASER dark photon search at LHC)

Also: 2506.13588, 2410.11367, 2412.03677, 2502.19817, 2503.01607, 2506.02450

$\epsilon = 10^{-5}$ (dark photon kinetic mixing parameter)

Dark photon A' $\rightarrow e+e-$ decay channel probed at FASER.

Vector boson mixing with SM photon provides vacuum coupling.

5. Integrated BSM Force Budget

$$F_{\text{quark}} = 1.54e7 \text{ N} \quad (\text{DOMINANT} - 99.9\% \text{ of BSM total})$$

$$F_{\text{ALP}} = 1.0e4 \text{ N}$$

$$F_{\text{neutrino}} = 1.0 \text{ N}$$

$$F_{\text{dark}} = 1.0 \text{ N}$$

$$F_{\text{BSM_total}} = 1.54e7 \text{ N} \quad (\text{approximately } F_{\text{quark}})$$

Compare to $F_{\text{LENR}} = 6.17e37 \text{ N}$:

$$F_{\text{BSM}} / F_{\text{LENR}} = 2.5e-31 \quad (\text{negligible vs LENR})$$

However: BSM terms connect UQFF to experimentally measurable particle physics parameters, providing validation pathways.

6. 8-System Recalculation

All 8 sonification systems recalculated with integrated BSM + F_{LENR} :

$$F_{\text{U_Bi_integrated}} = -F_0 + \text{momentum} + \text{gravity} + \rho_{\text{vac}} + F_{\text{LENR}} + F_{\text{BSM}}$$

BSM contribution: $+1.54e7 \text{ N}$ across all systems

Negligible vs F_{LENR} but establishes particle-cosmology bridge.

Conclusion

The 24-paper arXiv landscape yields four BSM $F_{\text{U_Bi_i}}$ terms. F_{quark} 's dominance at $1.54e7 \text{ N}$ reflects the CKM matrix's direct coupling to vacuum energy exchange. While negligible against F_{LENR} ($6.17e37 \text{ N}$), these terms provide experimentally measurable connections between UQFF and collider physics (Belle II, FASER, ALP searches).

Copyright - Daniel T. Murphy, daniel.murphy00@gmail.com, created by Davinci-SuperGrok, analyzed by Grok 3, and SuperGrok, created by xAI, dated June 19-20, 2025, EDT, location 41.0997 N, 80.6495 W (Youngstown, OH, USA).